College of Engineering **Renewable Energy Program**

University of Ha'il

Economics of Energy Applications
REE 403

Course Project

What are the impacts of PV panel degradation rate, initial system efficiency, interest rate, and PV project location on the LCOE of the Saudi Vision 2030 PV goal?

Names: Hakeem Faisal Almohawes
Salman Khaled Alwajaan
Khaled Bader Aljabbari

IDs: 201903115
201904914
201902861

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Abstract

This study investigates the factors influencing the levelized cost of energy (LCOE) of solar photovoltaic (PV) systems in Saudi Arabia, focusing on the country's ambitious renewable-energy targets under Saudi Vision 2030. The key parameters analysed are initial system efficiency, panel degradation rate, interest rate, and PV project location. The study shows that higher initial efficiency and lower degradation rates significantly reduce the LCOE, while higher interest rates tend to increase energy costs. The results highlight the importance of technological advances — such as improving PV panel efficiency and extending system lifetimes — together with favourable financing conditions, in making solar energy more economically competitive. This work contributes to the ongoing effort to diversify Saudi Arabia's energy mix and provides useful insight into the economic feasibility of large-scale solar projects. Future research can examine site-specific solutions based on local conditions and study the role of energy storage and policy incentives in further reducing the LCOE.

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Chapter 1

Introduction

The global energy sector is witnessing a significant transition towards sustainable and renewable energy sources, driven by the urgent need to combat climate change, reduce greenhouse-gas emissions, and address the growing global demand for energy. In this context, Saudi Arabia, one of the world's largest producers of fossil fuels, has taken a strategic approach to diversifying its energy portfolio through its ambitious Vision 2030 initiative. This vision aims to reduce the Kingdom's dependence on oil by integrating renewable-energy sources, with solar energy playing a pivotal role in achieving this transformation. Saudi Arabia's geographical location within the Sun Belt, offering some of the highest levels of solar irradiance in the world, makes solar photovoltaic (PV) systems an ideal solution for meeting the Kingdom's energy-diversification goals.

However, the successful implementation of large-scale solar PV projects in Saudi Arabia comes with unique challenges. The harsh desert environment — characterized by high temperatures, intense sunlight, and frequent dust and sandstorms — can significantly impact the performance, efficiency, and lifespan of PV systems. These environmental factors not only accelerate panel degradation but also necessitate the adoption of advanced technologies and innovative solutions to maintain the efficiency and economic viability of solar projects over time.

A critical metric for assessing the feasibility of solar PV systems is the Levelized Cost of Energy (LCOE), which represents the average cost of electricity generated by a system over its lifetime. Achieving a competitive LCOE is vital for ensuring the long-term sustainability of solar projects. Several factors influence the LCOE of PV systems, including panel degradation rates, initial system efficiency, interest rates, and project location. Each of these factors plays a crucial role in determining the overall cost and energy output of a PV project. For example, lower degradation rates and higher initial system efficiencies can improve energy production and reduce costs, while favourable interest rates and strategic site selection can further enhance project economics.

The Saudi government has introduced several supportive policies and financial incentives, such as subsidies and low-interest financing, to encourage investment in solar energy and accelerate the adoption of renewable-energy technologies. These initiatives are expected to play a significant role in overcoming financial and technical barriers to solar deployment. Furthermore, Saudi Arabia's focus on developing regions like NEOM and Al Jouf as hubs for renewable energy demonstrates the Kingdom's commitment to leveraging its natural resources to achieve its Vision 2030 goals. This project aims to explore the factors influencing the LCOE of PV systems in Saudi Arabia and provide insight into how these factors can be optimized to ensure the economic viability of solar energy projects.

1.1 Research Background

Saudi Arabia's Vision 2030 is an innovative strategy aimed at broadening the country's energy sources, with a specific focus on solar energy within its renewable-energy plan. Ensuring a competitive LCOE is crucial for the economic viability of photovoltaic (PV) projects in this perspective. Various elements have a significant impact on the LCOE, such as the degradation rate of PV panels, the efficiency of the system at the beginning, interest rates, and the location of the project. The decline in energy-conversion efficiency over time, known as the degradation rate of PV panels, directly affects energy output and project economics. Research suggests that degradation rates usually fall within the range of 0.5% to 1% per year; nevertheless, in Saudi Arabia's severe desert weather, marked by intense heat and dust, these rates can increase unless robust, climate-appropriate PV technologies are utilized (Jordan & Kurtz, 2013). Moreover, the efficiency of the system at the beginning, which depends on the quality of the PV modules, inverters, and overall system design, is crucial in lowering the LCOE. Systems with increased efficiency produce greater energy output per given area, leading to substantial cost reductions. With Saudi Arabia's high levels of solar radiation, improving system efficiency presents significant opportunities to lower costs.

Aside from technical variables, financial factors such as interest rates greatly impact the LCOE. Reduced interest rates decrease capital costs, making project financing more affordable. The Saudi government has made this easier by providing subsidies and favourable financing to encourage private-sector investment in solar energy (Lazard, 2023). Finally, the LCOE of PV projects is affected by differences in solar-resource availability, land costs, and distance to transmission infrastructure, depending on the geographical location. Saudi Arabia's location in the Sun Belt, especially areas such as NEOM and Al Jouf, offers significant potential for solar energy. Yet it remains crucial to choose the best site and connect it efficiently to the national grid in order to reduce transmission losses.

1.2 Research Motivation

The motivation for this study stems from the increasing worldwide need for sustainable-energy solutions and Saudi Arabia's dedicated focus on renewable energy as outlined in Vision 2030. As one of the leading oil producers, the Kingdom is looking to expand its energy sources to lessen reliance on fossil fuels and guarantee sustainable energy security. Solar energy is crucial in reaching these objectives due to its vast potential in Saudi Arabia. Nevertheless, for solar PV systems to become economically feasible and widely embraced, the LCOE needs to be reduced. Addressing important factors such as PV panel degradation, system efficiency, financing conditions, and ideal project locations is necessary for this.

Saudi Arabia's distinctive weather conditions — such as elevated temperatures, dust, and occasional sandstorms — create substantial obstacles for the effectiveness and durability of PV systems. These environmental conditions speed up panel deterioration, impacting the overall energy production and financial viability of the system. It is essential to comprehend these effects and create plans to reduce them in order to guarantee the prosperity of solar initiatives in the region.

Additionally, the competitiveness of solar-energy projects is directly impacted by the initial efficiency of the system and financing conditions. Government-led programs provide subsidies and low-interest financing options, encouraging the exploration of financial models to further decrease the LCOE of solar systems. Furthermore, taking advantage of Saudi Arabia's geographical benefits, such as the abundant solar energy in areas like NEOM and Al Jouf, could optimize energy generation and reduce expenses.

This study is driven by the necessity to connect technical, environmental, and financial factors in solar PV projects. It aims to provide practical insights that support Vision 2030's renewable-energy goals and the global shift towards sustainable energy systems by examining the interaction of these factors.

1.3 Research Outline

- Saudi Vision 2030 goal
- PV panel degradation rate
- Initial system efficiency
- Interest rates
- PV project location
- Levelized Cost of Electricity (LCOE)

Chapter 2

Literature Review

2.1 Saudi Vision 2030

The Saudi Vision 2030 framework aims to diversify the Kingdom's economy and decrease its dependence on oil by enhancing innovation and sustainability. Renewable energy is a keystone of this plan, with specific targets for solar power — including 40 GW of photovoltaic capacity by 2030. It also aligns with the broader global sustainability objectives of reducing carbon emissions and fighting climate change. Saudi Arabia has already made considerable progress, with multiple solar projects such as those described below.

2.1.1 Sakaka PV

The Sakaka Solar Power Plant in Al Jouf, overseen by the Ministry of Energy, was the first renewable-energy project launched under the patronage of King Salman bin Abdulaziz Al Saud. The plant uses photovoltaic (PV) technology to generate electricity. It consists of 1.2 million solar panels arranged over 6 km^2 of land, and the site was carefully selected by a specialized Saudi technical team to ensure the highest possible quality of electrical-power production. The plant contributes to the reduction of 606,000 tons of carbon emissions, has a production capacity of 300 MW, and covers 44,000 housing units. The project set a world record for low cost in the solar PV sector at the time, at 0.08775 SAR per kilowatt-hour [1].

Figure 2.1. Sakaka PV IPP.

2.1.2 Sudair Solar PV

Located in Saudi Arabia's Riyadh region, Sudair is one of the largest and most significant solar initiatives under the Kingdom's Vision 2030 plan. With a capacity of 1.5 GW, Sudair is set to be a key contributor to Saudi Arabia's renewable-energy targets, expected to generate enough power to supply electricity to 185,000 homes and offset 2.9 million tons of CO_2 emissions annually. The project recorded one of the lowest levelized costs of electricity in the world at 1.239 US cents per kWh [2].

Figure 2.2. Sudair PV IPP.

2.1.3 Layla Solar PV Plant

The Layla Solar PV Plant is a pivotal part of Saudi Arabia's renewable-energy expansion under the National Renewable Energy Program (NREP). Located in the Riyadh region, the project is designed to produce about 80 MW of electricity, contributing to the country's goals of reducing dependency on fossil fuels. By utilizing solar photovoltaic technology, it aligns with Vision 2030's aim of developing a sustainable energy infrastructure while minimizing environmental impact [3].

Figure 2.3. Layla Solar PV IPP.

2.1.4 Rabigh Solar Project

Situated in the western region of Saudi Arabia, the Rabigh Solar Project focuses on leveraging the Kingdom's high solar irradiance. This project is an integral part of the national strategy to achieve the 40 GW renewable-energy target by 2030. It highlights Saudi Arabia's commitment to transitioning to clean energy while fostering private-sector involvement [4].

2.1.5 Solar PV Cell & Module Manufacturing Plant

Saudi Arabia is also investing in localized solar-energy technology through the Solar PV Cell & Module Manufacturing Plant. Established by the King Abdulaziz City for Science and Technology (KACST), the Solar PV Cell & Module Manufacturing Plant and PV Reliability Laboratory produce solar panels and cells. The facilities bring the latest solar-energy technologies to Saudi Arabia and aim to create top-quality equipment that can withstand extreme heat and sandstorms [5].

Figure 2.4. Solar PV Cell & Module Manufacturing Plant.

2.2 PV Panel Degradation Rate

The degradation rate of a photovoltaic (PV) panel refers to the gradual decline in its ability to convert sunlight into electricity over time. This degradation is a natural process caused by environmental exposure, material wear, and operational factors. Typically, modern PV panels degrade at an average rate of 0.5% per year, meaning that after 25 years most panels retain approximately 87.5% of their original efficiency. Understanding degradation is essential for predicting the performance, financial viability, and sustainability of PV systems over their lifetime. Some companies, such as SunPower, guarantee 98% production for the first year and degradation of no more than 0.25% per year for the following 24 years; the warranty guarantees that the power output at the end of the 25th year will be at least 92% of the guaranteed peak-power rating [6].

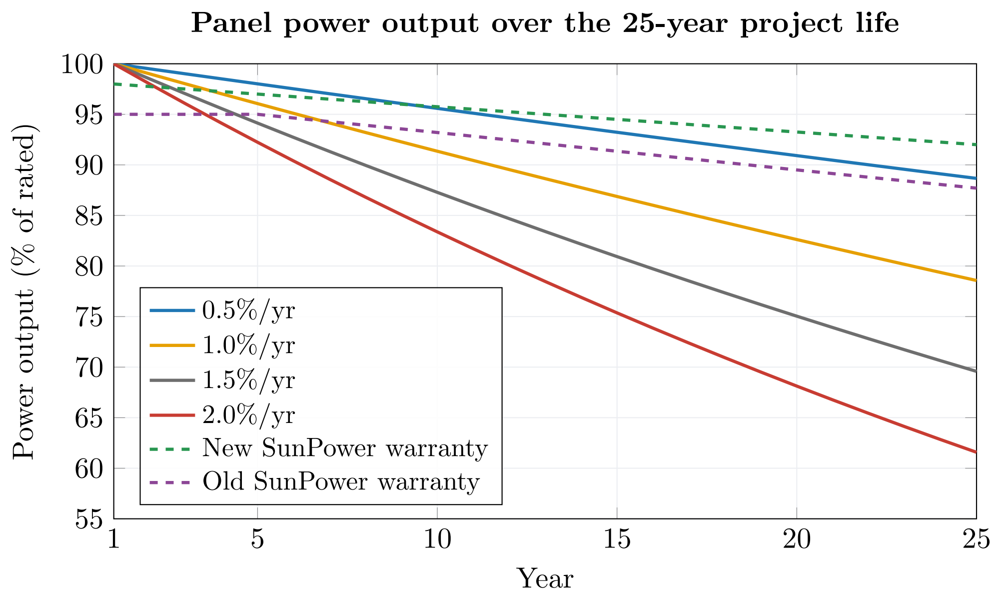

Figure 2.5. Panel degradation.

2.2.1 Causes of Degradation

PV panel degradation occurs due to both intrinsic and extrinsic factors. **Intrinsic degradation** includes material aging and microcrack formation in silicon cells, while **extrinsic factors** include environmental stressors such as UV radiation, thermal cycling, and moisture ingress. These stressors accelerate the breakdown of encapsulants, frame materials, and anti-reflective coatings. Manufacturers combat these issues by employing advanced materials and protective technologies, but no panel is entirely immune to degradation [7].

2.2.2 Impact of Degradation on Output

As panels degrade, their ability to produce electricity diminishes, reducing energy output and impacting financial returns. This reduction directly affects metrics such as the Levelized Cost of Electricity (LCOE) and Return on Investment (ROI). For example, panels with higher degradation rates may fail to meet production guarantees, requiring early replacement. Accurate modelling of degradation is therefore crucial for long-term project planning and financing [8].

2.2.3 Importance of Initial Quality

The initial build quality of PV panels plays a significant role in degradation rates. High-quality panels from reputable manufacturers often exhibit lower and more predictable degradation rates due to rigorous testing, better materials, and advanced engineering. Certification standards, such as those from IEC and UL, ensure that panels meet reliability benchmarks before deployment [9].

2.2.4 Climatic and Environmental Conditions

The operating environment significantly influences degradation. In hot, humid, or high-UV regions, panels are exposed to more severe stress, leading to faster degradation. For instance, desert climates can cause increased wear on anti-reflective coatings and encapsulants. Selecting climate-optimized panels and employing protective measures such as proper mounting and cooling systems can mitigate these effects. The biggest cause of PV panel degradation in Saudi Arabia is **dust accumulation** due to frequent dust storms and the high dust intensity of desert environments. Dust deposition reduces sunlight reaching the panels, leading to efficiency losses of up to 20–40% if the panels are not cleaned regularly. Additionally, **high temperatures** accelerate thermal degradation and material aging, while **UV exposure** contributes to encapsulant and anti-reflective-coating wear. These factors collectively lead to higher degradation rates compared with more temperate climates [10].

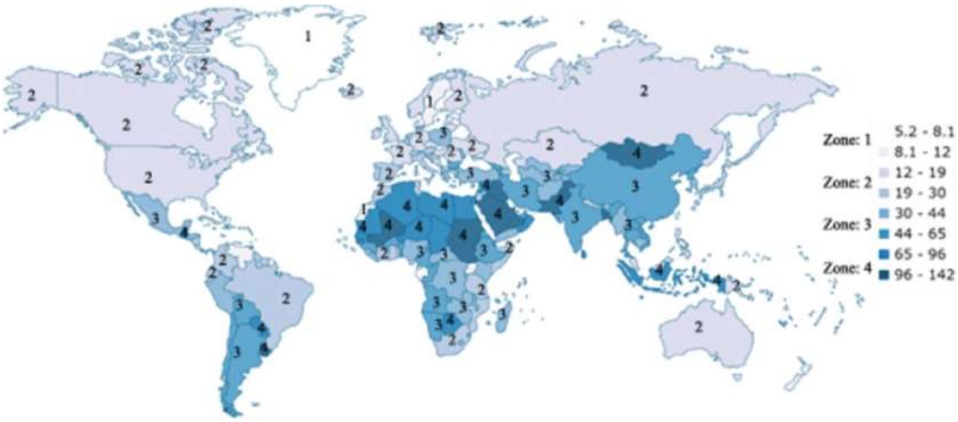

Figure 2.6. Dust intensity around the world.

2.2.5 Advances in PV Technology to Reduce Degradation

Innovations in PV panel technology aim to minimize degradation rates. Improvements include the use of monocrystalline over polycrystalline silicon, better encapsulation materials, and advanced coatings that resist environmental damage. Some manufacturers now offer panels with degradation rates as low as 0.2% per year, extending the productive life of installations beyond 30 years [11].

2.3 Initial System Efficiency

The initial system efficiency of a PV system plays a significant role in determining how degradation will affect its long-term performance. A system with higher initial efficiency typically has a higher power output, which makes it more resilient to the impacts of degradation over time. As degradation mechanisms such as **Light-Induced Degradation (LID)** and **Potential-Induced Degradation (PID)** cause a decrease in panel performance, systems with a higher starting efficiency maintain better overall output for a longer period [11]. Several factors related to initial system efficiency affect degradation.

2.3.1 Panel Type and Quality

Panel type and quality are crucial elements influencing the initial system efficiency of a PV system and how it performs over time. Different types of panels — such as **monocrystalline**,

polycrystalline, and **thin-film** — exhibit varied levels of efficiency and degradation behaviour, which directly affect the long-term performance of a PV system [12].

Monocrystalline panels: Known for their high efficiency, monocrystalline silicon panels typically have higher initial system efficiency compared with other types. Their efficiency typically ranges from 18% to 22%, and they generally experience lower degradation rates over time. The higher purity silicon used in monocrystalline panels ensures better performance; even as degradation factors such as LID and PID occur, these panels tend to perform better under high temperatures and low-light conditions, further extending their efficiency over their lifespan [11].

Polycrystalline panels: Polycrystalline panels are generally less efficient than monocrystalline panels, with initial efficiencies between 15% and 18%. While they are more affordable, they also tend to have a higher rate of degradation compared with monocrystalline panels, particularly when exposed to high temperatures or prolonged UV radiation. Over time, this higher degradation rate leads to a greater loss in performance [11].

Thin-film: Thin-film technology, including amorphous silicon and cadmium telluride (CdTe), offers lower initial efficiency (around 10% to 12%) but can be beneficial in certain applications due to its flexibility and lower production costs. However, thin-film panels typically degrade faster due to their susceptibility to factors such as moisture ingress and UV radiation. They may also experience light-induced degradation more severely in their early years of operation, leading to a quicker decline in energy output [11].

Figure 2.7. Different types of panels.

2.3.2 Manufacturing Quality

Manufacturing quality plays a significant role in determining the initial system efficiency of a PV system. The precision and methods used in manufacturing directly affect the panel's performance, lifespan, and degradation over time. High-quality panels often exhibit lower degradation rates and maintain their efficiency for longer periods.

A. **Cell efficiency:** The manufacturing process affects the efficiency of the solar cells. Advanced technologies such as **Passivated Emitter and Rear Contact (PERC)** and **n-type silicon** cells enhance the initial efficiency of PV panels by reducing energy losses caused by surface recombination. These innovations result in higher energy-conversion rates, especially in monocrystalline panels, and mitigate performance drops over time [13].

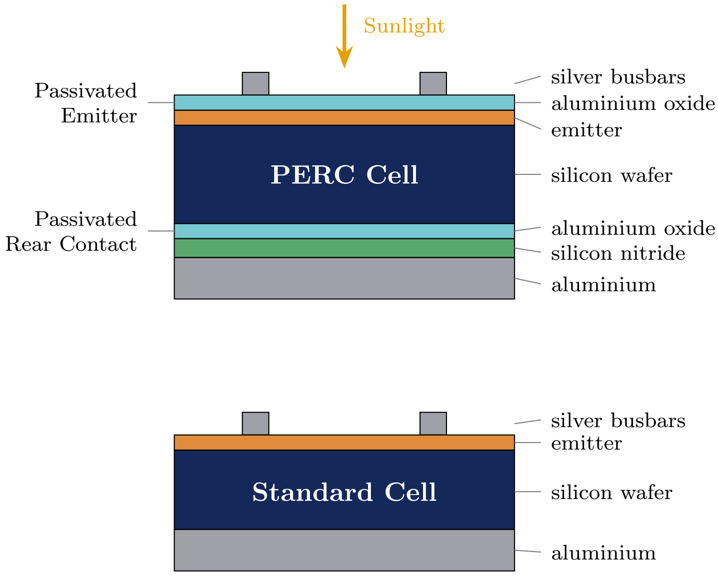

Figure 2.8. PERC cell.

B. **Encapsulation and coating quality:** The quality of encapsulation materials (such as EVA or PVB) and protective coatings plays a crucial role in preventing moisture ingress, UV degradation, and thermal stress. Panels manufactured with superior materials are more resistant to delamination and microcracks, which are common causes of degradation in lower-quality panels [14].

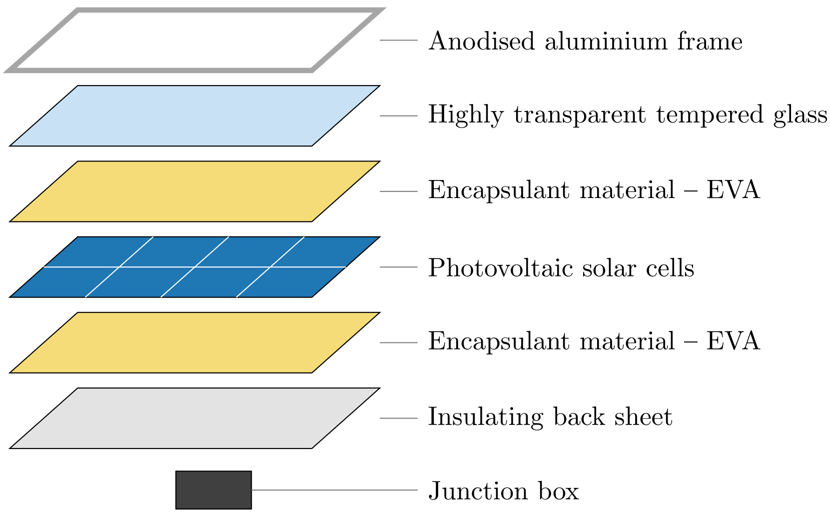

Figure 2.9. EVA in a solar cell.

C. **Manufacturing standards and certification:** Panels produced under strict international standards (such as IEC 61215 or UL 1703) are tested for performance, durability, and safety. Panels with higher manufacturing standards are less prone to defects and fail less often in real-world conditions, leading to lower initial degradation rates. Furthermore, panels that undergo rigorous quality-control measures during production tend to exhibit higher consistency in performance [15].

D. **Material purity:** The use of high-purity silicon and other high-quality materials results in panels with fewer defects in the crystal structure, leading to better energy output and less degradation over time. Lower-quality materials can lead to a higher incidence of **microcracks**, which gradually reduce the panel's efficiency [7].

2.3.3 Temperature Coefficients

Temperature is a critical factor that affects the initial system efficiency of photovoltaic (PV) panels. The temperature coefficient of a solar panel indicates how much its efficiency decreases as the temperature increases. This is particularly important for solar panels used in hot climates, where high temperatures can significantly reduce performance over time. It quantifies the decrease in power output (in percentage terms) for every degree Celsius increase in temperature above 25 °C (the standard test condition). For example, a typical temperature coefficient for monocrystalline panels ranges from −0.3% to −0.4% per degree Celsius, while polycrystalline panels often have a coefficient of around −0.4% to −0.5% per degree Celsius [16].

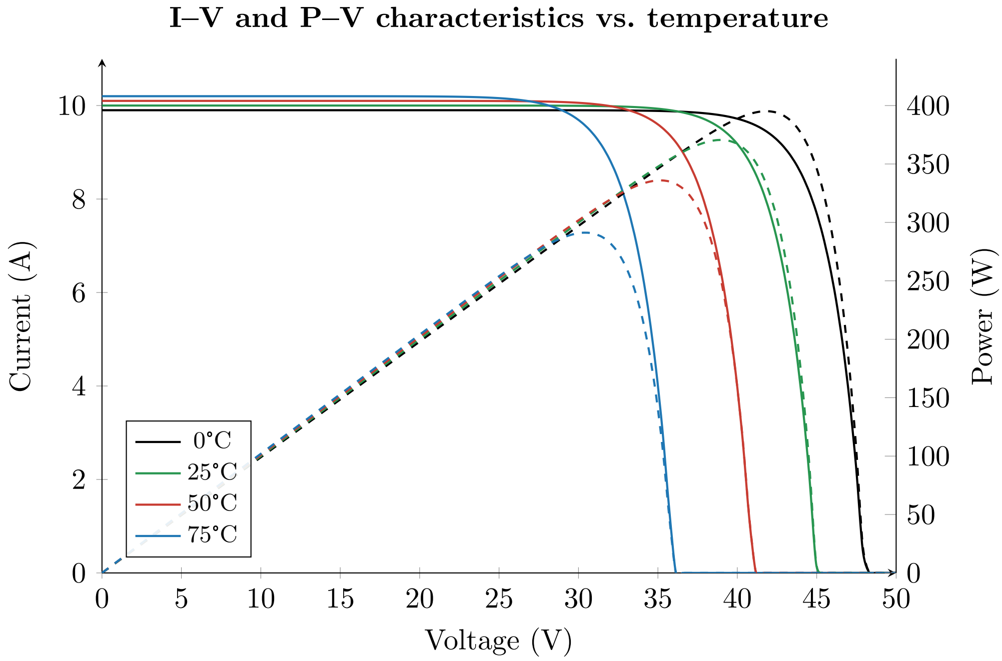

Figure 2.10. I–V and P–V characteristics of solar cells.

2.3.4 Design and Installation

The design and installation of photovoltaic (PV) systems significantly impact their initial efficiency. Proper panel orientation and tilt are crucial for maximizing sunlight capture, while avoiding shading from trees or buildings prevents power loss. Adequate spacing between panels ensures proper air circulation to prevent overheating, which would otherwise reduce efficiency due to higher temperature coefficients. Additionally, the quality of wiring and inverter systems directly affects energy conversion, and professional installation helps avoid issues such as microcracks and electrical faults. Overall, optimized design and careful installation can enhance performance and longevity, reducing the risk of early degradation [17].

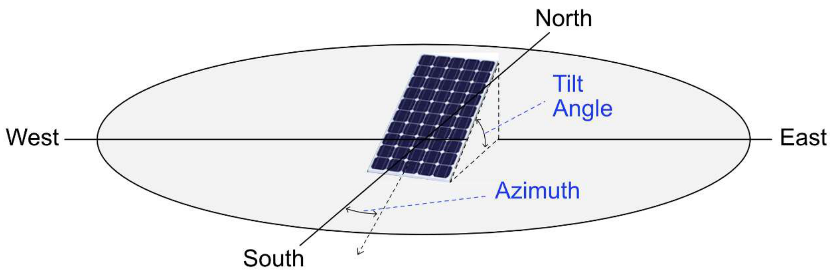

Figure 2.11. Solar angles.

2.4 Interest Rate

The interest rate is the cost of borrowing money or the return on investments, expressed as a percentage of the principal amount over time, and it plays a significant role in the financing and economic viability of photovoltaic (PV) projects. In the context of PV systems, interest rates determine the cost of capital for developers, influencing upfront investment requirements and long-term project cash flows. Lower interest rates reduce financing costs, enabling more competitive energy pricing and improving the attractiveness of solar projects. Conversely, high interest rates increase the LCOE by raising the expense associated with loans or other forms of debt financing. Interest rates also affect the discount rate used in LCOE calculations, impacting the valuation of future energy production and operational expenses. Thus, managing interest rates through favourable financing structures, such as green bonds or subsidized loans, is critical for achieving cost-efficient solar-energy deployment [18].

Figure 2.12. Saudi Arabia's short-term interest rate from Jan 2007 to Aug 2024.

2.5 PV Location

The location of a photovoltaic (PV) system is a decisive factor in its performance and economic viability. Solar irradiance, which varies by location, is the primary driver of energy output. Areas with high solar irradiance, such as those in Saudi Arabia, benefit from increased electricity-generation potential, leading to lower LCOE. However, location-specific challenges such as extreme temperatures and environmental conditions must also be considered [19]. In desert regions, for instance, high ambient temperatures can negatively impact PV efficiency due to thermal losses. A study conducted by KAUST in Saudi Arabia highlighted that modules operating in high-temperature environments regularly reach 60–65 °C, reducing power output and potentially shortening their lifespan.

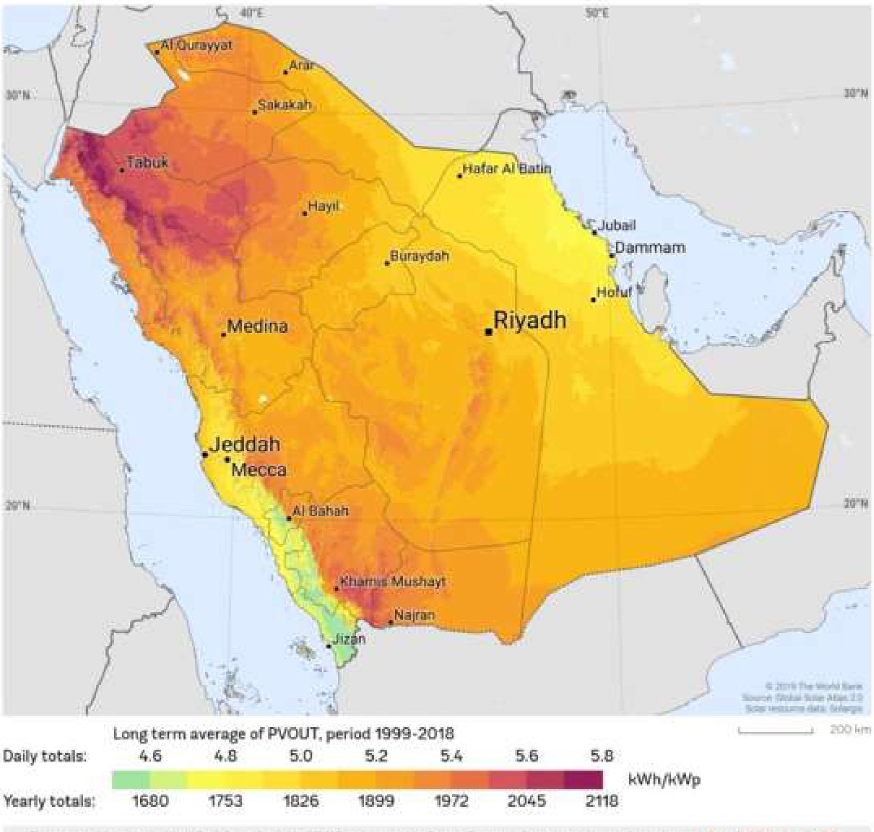

Figure 2.13. Photovoltaic electricity potential in Saudi Arabia.

2.6 Levelized Cost of Electricity (LCOE)

The Levelized Cost of Electricity (LCOE) is a widely used metric to assess the economic viability of energy-generating technologies, including photovoltaic (PV) systems. LCOE represents the average cost per kilowatt-hour (kWh) of electricity over the lifetime of an energy system, accounting for all costs — including initial capital, operations and maintenance (O&M), and end-of-life costs — discounted to present value. Studies show a significant decline in PV LCOE globally, driven by advances in technology, economies of scale, and supportive policies. For instance, between 2010 and 2020, global average PV LCOE dropped by 82%, with further reductions in regions like Saudi Arabia, where abundant solar resources and competitive bidding under Vision 2030 yielded record-low costs. Key factors influencing LCOE include degradation rates, which affect long-term performance, and discount rates, which vary by region and impact financing costs. Additionally, assumptions about system lifetime — such as operational spans of 20 versus 30 years — significantly alter LCOE outcomes [20].

Chapter 3

Research Methodology

In this section we present the equations used to assess the PV project. The Levelized Cost of Electricity is defined as

$$\mathrm{LCOE} = \frac{I_0 + \sum_{t=0}^{n} \mathrm{O\&M}_t}{\sum_{t=0}^{n} E_t}, \tag{3.1}$$

where I_0 is the initial investment, $\mathrm{O\&M}_t$ is the operations-and-maintenance cost during a single period t, n is the number of periods corresponding to the project's lifetime, and E_t is the total energy generated during period t.

However, this equation does not consider the interest paid on the initial investment, so we adjust it to take the interest rate into account:

$$\mathrm{LCOE} = \frac{I_0 + I_{\mathrm{int}} + \sum_{t=0}^{n} \mathrm{O\&M}_t}{\sum_{t=0}^{n} E_t}, \qquad I_{\mathrm{int}} = I_0\left[(1+r)^n - 1\right], \tag{3.2}$$

where r is the annual interest rate and I_{int} is the total financing cost over the project life. The energy generated in year t is

$$E_t = P_{\mathrm{rating}} \times \mathrm{PSH} \times 365 \times \eta \times (1 - \mathrm{DR})^t, \tag{3.3}$$

where P_{rating} is the solar-system power rating, PSH is the average peak sun hours per day, η is the initial system efficiency, and DR is the annual degradation rate.

Since we are studying the impacts of PV panel degradation rate, initial system efficiency, interest rate, and PV project location on the LCOE of the Saudi Vision 2030 PV goal, we rely on available data from completed solar projects in Saudi Arabia (Table 3.1).

Table 3.1. Initial investment cost of completed Saudi PV projects.

Project	Capacity	Initial investment cost ($/W)
Sakaka PV	300 MW	$1.01/W
Al-Shuaibah	2660 MW	$0.91/W
Sudair PV	1500 MW	$0.60/W
Layla Solar PV	80 MW	≈$0.64/W
Average	—	**$0.79/W**

We use the average of $0.79/W as the initial investment cost in the calculations. The solar-system power rating is taken as 40 GW, the capacity Saudi Arabia aims to achieve under Vision 2030. The global horizontal irradiance (GHI) in Saudi Arabia varies by region, with annual averages typically ranging between 5.5 and 6.5 peak sun hours per day, so we assume 6 peak sun hours per day for the calculations. Operations-and- maintenance (O&M) costs for PV projects in Saudi Arabia are relatively low, reflecting the country's favourable solar conditions and the efficiency of its utility-scale plants; industry estimates suggest that O&M costs for large-scale PV plants in the region average around $10 per kilowatt (kW) per year, which is the value assumed here. Finally, a project lifetime of $n = 25$ years is assumed, consistent with the standard performance warranty of modern PV modules.

Table 3.2. Baseline assumptions used in the LCOE calculations.

Parameter	Value
Initial investment cost	$0.79/W
Solar-system power rating	40 GW
Average peak sun hours	6 h/day
O&M cost	$10/kW/year
Project lifetime, n	25 years

Chapter 4

Results & Discussion

Because the LCOE is an intensive quantity (cost per unit of energy), it is independent of the chosen system size; the 40 GW rating therefore cancels out and the results depend only on the per-watt cost, the peak sun hours, the O&M cost, the lifetime, and the three parameters swept below. In each case one parameter is varied while the others are held at their reference values.

4.1 Interest rate = 0, initial efficiency = 1, varying degradation rate

This chart illustrates that the LCOE *increases* as the degradation rate increases. A higher degradation rate means the system loses generating capacity more quickly over time, producing less total energy over its lifetime, which raises the cost per kilowatt-hour. This highlights the importance of using high-quality materials and proper maintenance to minimize system degradation and preserve long-term performance.

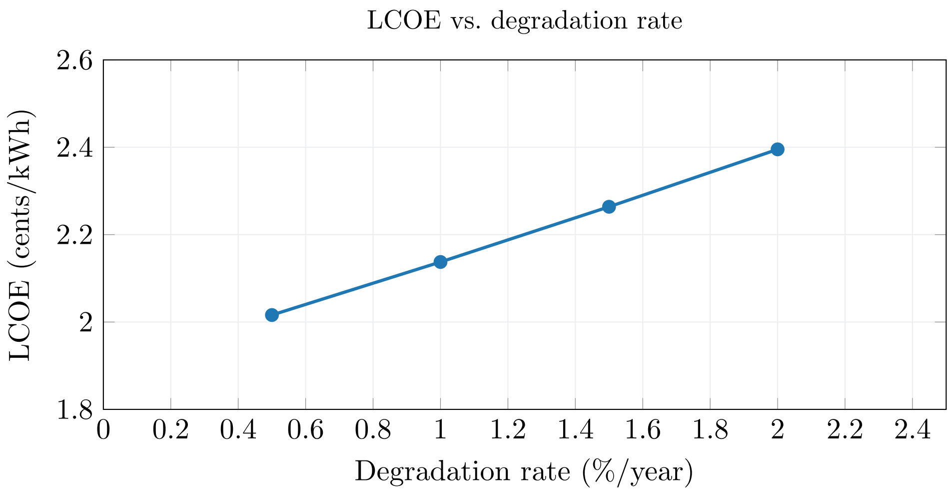

Figure 4.1. LCOE vs. degradation rate.

4.2 Degradation rate = 0, initial efficiency = 1, varying interest rate

This chart shows that the LCOE *rises* as the interest rate increases. Higher interest rates result in greater financing costs for the project, which directly increase the overall cost of electricity generation. This emphasizes the critical role of securing low-interest loans to reduce the financial burden on the project and enhance its cost-effectiveness over its lifetime.

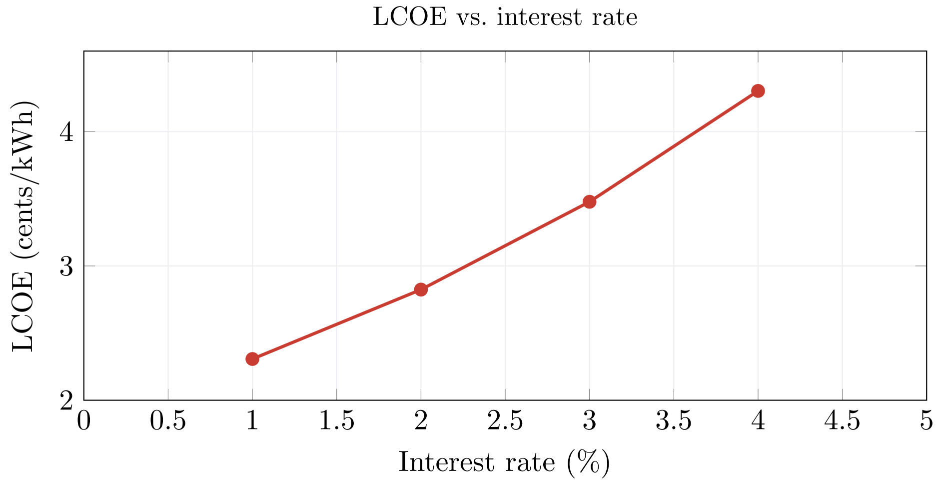

Figure 4.2. LCOE vs. interest rate.

4.3 Degradation rate = 0, interest rate = 0, varying initial efficiency

This chart demonstrates that the LCOE *decreases* as the initial system efficiency increases. Higher- efficiency systems generate more electricity for the same installed capacity, reducing the cost per kilowatt-hour of energy produced. This highlights the importance of optimizing system efficiency — through high-performance PV panels and advanced system design — to improve the economic viability of solar projects.

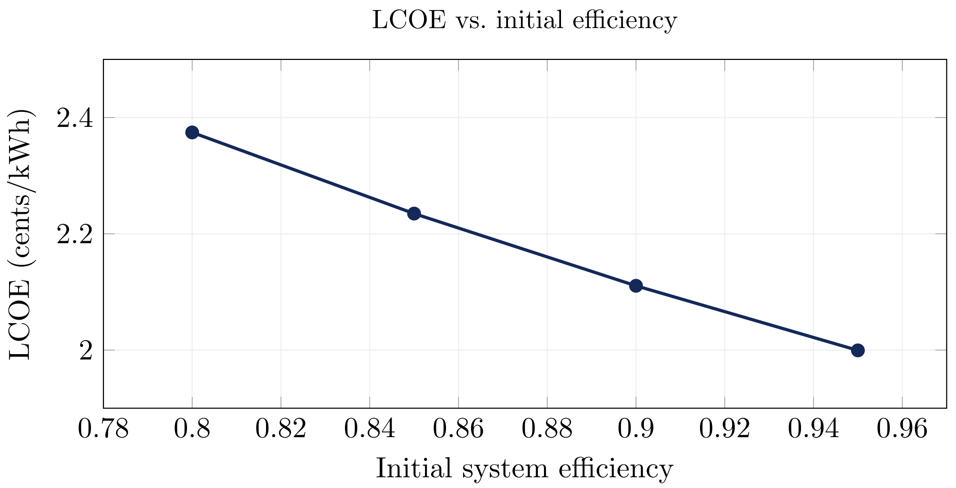

Figure 4.3. LCOE vs. initial efficiency.

4.4 Discussion

The findings show that the LCOE is greatly affected by key system factors such as starting efficiency, interest rate, and degradation rate. Higher initial system efficiency leads to a reduced LCOE, because more electricity is generated for the same capacity, allowing the fixed costs to be distributed over a larger energy output. This underscores the significance of using high-performance PV panels and advanced system designs. Conversely, when the interest rate goes up, the LCOE also goes up considerably, because the increased financing expenses raise the overall project cost — highlighting the importance of obtaining low-interest loans or incentives to ensure the economic viability of solar projects. Furthermore, the speed at which degradation occurs is crucial, as a faster rate hastens the decline in energy generation over time, ultimately decreasing the total energy output over the system's lifetime and raising the LCOE.

Across the scenarios examined, the computed LCOE ranges from about 1.9 to 4.3 cents/kWh. Reassuringly, these values fall in the same range as the record-low tariffs achieved by actual Saudi projects — for example, Sakaka ($\approx$ 2.3 cents/kWh) and Sudair ($\approx$ 1.24 cents/kWh) — which supports the validity of the model and its assumptions. In general, improving efficiency, obtaining good financing terms, and reducing degradation are all important for reaching the lowest LCOE and achieving cost-effective, sustainable solar-energy deployment.

Chapter 5

Conclusion & Future Work

In conclusion, this project uncovered the key factors that influence the levelized cost of electricity (LCOE) of a photovoltaic (PV) system and provided a comprehensive analysis of how variables such as initial system efficiency, degradation rate, and financial parameters such as interest rates affect the economic feasibility of a solar-energy project. The results highlight the importance of improving solar-panel efficiency and minimizing degradation rates over time to achieve reduced energy costs and improved financial sustainability. Additionally, obtaining favourable financial terms, such as low-interest loans, can significantly reduce the LCOE, making solar energy a more competitive and viable alternative. These findings contribute to the broader goals of Saudi Vision 2030, which emphasizes the role of cost-effective, sustainable renewable energy in reducing reliance on fossil fuels and helping to diversify the economy.

Looking to the future, there are significant opportunities for further work. Research efforts can focus on advancing photovoltaic technology to achieve higher efficiencies and lower degradation rates, thereby extending system life and improving overall performance. Integrating energy-storage systems with solar PV, such as battery technologies, can be explored to improve energy reliability and reduce reliance on the electricity grid. Additionally, further research on the impact of regional climatic conditions — including high temperatures and dust accumulation — on PV system performance could provide solutions better suited to Saudi Arabia's unique environment. Policy research aimed at developing effective incentives, such as subsidies, feed-in tariffs, and tax breaks, will continue to foster the adoption of renewable-energy systems. By addressing these areas, future research can build on this work and accelerate the transition to a sustainable energy future while aligning with Saudi Arabia's ambitious renewable-energy goals.

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